Linux Fundamentals
Course summary | Instructor: Beau Carnes

A short summary of the course: what Linux is, how to move around the command line, and how files, editors and permissions work.

Week 1-2
What is Linux
Linux is an open-source kernel created by Linus Torvalds in 1991. Together with GNU tools it forms a full operating system used on servers, phones (Android), embedded devices and supercomputers.

Distributions and Installation
A distribution bundles the kernel with software and a package manager. Popular ones are Ubuntu, Debian, Fedora and Arch. You usually install from a bootable USB image.

The Graphical Interface
Desktop environments such as GNOME, KDE Plasma and Xfce provide windows, menus and file managers. They are optional: servers usually run without any graphical interface.

The Linux Philosophy
Programs should do one thing well. Everything is treated as a file, text is the common interface, and small programs are combined with pipes (|).

Week 3-4
Basic Commands
Use pwd to print the current directory, ls to list files, cd to move between directories and man to read a command’s manual. Options such as ls -l change the output.

Working with Files
Create with touch and mkdir, copy with cp, move or rename with mv, delete with rm. rm -r removes directories and cannot be undone. Read files with cat or less.

Text Editors
nano is simple and shows its shortcuts on screen. vim is modal: press i to insert text, Esc to return to normal mode, and :wq to save and quit (:q! quits without saving).

Users and Permissions
Every file has an owner, a group and permissions for read, write and execute. chmod 755 file gives the owner full access and others read and execute. sudo runs a command as the superuser.

Quick reference

pwd	Print the working directory
ls -la	List all files with details
cd ..	Go up one directory
chmod +x file	Make a file executable

grep text file Search for text in a file